Royal University of Phnom Penh
Faculty of Engineering
Department of Information Technology Engineering
Bachelor of Engineering in Data Science and Engineering

Course Specification

PART 1: VISION, MISSION, GOALS, AND OBJECTIVES

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<tr>
<td>PROGRAM VISION:
To be a leading program in Data Science and Engineering in Cambodia with regional recognition for excellence in education, research, and innovation that drives digital transformation and societal development</td>
<td>PROGRAM MISSION
Mission 1: Empower students with strong foundations in data science, engineering, and digital technologies, along with critical thinking, ethics, and problem-solving skills
Mission 2: Advance knowledge and innovation through high-quality research in data science, artificial intelligence, and emerging technologies that address local and regional challenges
Mission 3: Engage with industry and society by applying data-driven solutions to real-world problems, fostering collaboration, and contributing to Cambodia’s digital economy and sustainable development</td>
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<td>PROGRAM GOALS
Our program aims to:
Develop skilled graduates with strong foundations in data science and engineering.
Foster innovation, research, and problem-solving in AI and emerging technologies.
Promote ethical, responsible, and professional practice.
Strengthen collaboration with industry and society for digital transformation.
Cultivate leadership, adaptability, and lifelong learning.</td>
<td>PROGRAM EDUCATIONAL PHILOSOPHY
EP1: Critical Thinking and Data-Driven Decision Making: Develop graduates who apply critical thinking, logical reasoning, and data-driven approaches to solve complex real-world problems and support informed decision-making.
EP2: Research, Innovation, and Computational Intelligence: Foster strong capacity in research, innovation, and the use of computational methods, AI, and data science techniques to generate new knowledge and solutions.
EP3: Professional Practice and Industry Readiness: Prepare graduates with strong technical competence, digital literacy, and professional skills to meet evolving industry and employability demands.
EP4: Societal and Regional Responsiveness: Equip graduates to design data-driven solutions that address societal challenges and respond to regional development needs in Cambodia and ASEAN.
EP5. Intercultural Competence and Responsible Citizenship: Promote intercultural understanding, teamwork, and responsible citizenship.
EP6: National Development, Sustainability, and Digital Transformation: Encourage graduates to contribute to national development, cultural and environmental sustainability, and digital transformation through innovative data science and engineering solutions.</td>
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Program Educational Objectives (PEOs)

What graduates are expected to achieve within 3–5 years of graduation**:**

- **PEO1: Professional Practice**: Graduates will become competent professionals in data science and engineering, applying digital and analytical skills in diverse industries.
- **PEO2: Innovation & Problem Solving**: Graduates will demonstrate innovation, critical thinking, and problem-solving skills to design and implement data-driven solutions to real-world challenges.
- **PEO3: Communication & Collaboration**: Graduates will effectively collaborate and communicate with interdisciplinary teams and stakeholders in both local and international contexts.
- **PEO4: Ethics & Social Responsibility**: Graduates will practice with professionalism, ethics, and a commitment to social and environmental responsibility.
- **PEO5: Lifelong Learning & Leadership**: Graduates will pursue continuous learning, certifications, or advanced degrees and assume leadership roles in the digital economy.

Program Learning Outcome (PLOs)

Our program has ten PLOs:

<table>
<tr><th colspan="2">CQF Learning Domains</th><th rowspan="2">PLO</th><th rowspan="2">Specific/Generic</th><th rowspan="2">Learning/Assessment Domains</th></tr>
<tr><th>Major Domain</th><th>Learning Domain</th></tr>
<tr><td>MD1: Knowledge</td><td>LD1: Knowledge</td><td>PLO1: Apply knowledge in data science and engineering to develop appropriate solutions for real-world problems.</td><td>Specific</td><td rowspan="2">Cognitive</td></tr>
<tr><td>MD2: Cognitive skills</td><td>LD2: Cognitive skills</td><td>PLO2: Analyze data-related problems using logical reasoning and systems thinking.</td><td>Specific</td></tr>
<tr><td>MD3: Psychomotor/Technical skills</td><td>LD3: Psychomotor/Technical skills</td><td>PLO3: Utilize data science tools and technologies to develop technical solutions for practical applications.</td><td>Specific</td><td>Psychomotor</td></tr>
<tr><td rowspan="4">MD4: Interpersonal skills and responsibility</td><td>LD4: Interpersonal skills</td><td>PLO4: Participate effectively in multicultural and multidisciplinary teams with intercultural competence and responsible citizenship.</td><td>Generic</td><td rowspan="5">Affective</td></tr>
<tr><td>LD5: Responsibility</td><td>PLO5: Demonstrate leadership, accountability, and lifelong learning in professional practice.</td><td>Generic</td></tr>
<tr><td>LD6: Entrepreneurial skills</td><td>PLO6: Develop innovative and entrepreneurial data-driven solutions that support national development and cultural sustainability in Cambodia and the ASEAN region.</td><td>Specific</td></tr>
<tr><td>LD7: Ethics and Professionalism</td><td>PLO7: Make ethical decisions that reflect professional responsibility and awareness of social, cultural and environmental impacts.</td><td>Generic</td></tr>
<tr><td rowspan="3">MD5: Communication, information technology, and numerical skills</td><td>LD8: Communication</td><td>PLO8: Communicate ideas and findings clearly through oral, written, and visual form.</td><td>Generic</td></tr>
<tr><td>LD9: Information technology skills</td><td>PLO9: Utilize digital technologies and platforms to support communication, collaboration, and data-driven work.</td><td>Specific</td><td>Psychomotor</td></tr>
<tr><td>LD10: Numerical skills</td><td>PLO10: Apply mathematical, logical, and statistical reasoning in data analysis and problem solving.</td><td>Specific</td><td>Cognitive or Psychomotor</td></tr>
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Specific (Subject-Specific) PLOs: Directly related to data science and engineering knowledge, tools, and technical skills)
Generic PLOs: Transferable skills applicable across disciplines and professions

PART 2: COURSE DETAILS

Course Information

1. **Programme Title**	Bachelor of Engineering in Data Science and Engineering		
2. **Course Title**	Predictive Analytics		
3. **Course Code**	PAN202	4. **No. of Credits**	3
5. **Pre-requisites (If any)**	Math I–III; Statistics I–II; Advanced Programming (Python & OOP); Data Manipulation and Visualization		
6. **Course Instructor**	Chim Seyha	7. **Qualification**	Master’s degree in computer science
8. **Email**	chim.seyha@rupp.edu.kh	9. **Telephone No.**	096 5321 532
10. **Other Course Lecturer(s) (If any)**			
11. **Course Type**	Basic ☐ Core ☑ Elective ☐ Specialization ☐ MoEYS / HEIP ☐		
12. **Course Availability**	1st Semester ☐ 2nd` Semester ☑	**Year**	2
13. **Course Description/ Synopsis**	Predictive Analytics introduces students to techniques used to analyze historical data and build predictive models for decision making. The course focuses on practical skills in data preparation, exploratory data analysis, feature engineering, and predictive modeling using Python. Students will learn how to implement predictive analytics workflows using common data science tools such as **Pandas, NumPy, Matplotlib, and Scikit-learn**. The course covers regression models, classification models, and model evaluation techniques including Mean Squared Error (MSE), Accuracy, Precision, Recall, and ROC/AUC. A project-based learning approach is emphasized throughout the semester. Students progressively develop a **mini predictive analytics project using real-world datasets from Kaggle**, allowing them to experience the full data science workflow from data preprocessing to model evaluation and communication of analytical findings. By the end of the course, students will be able to build predictive models, evaluate their performance, and communicate insights derived from data effectively.		

14. **Course Learning Outcomes**

Here are the CLOs of this course:

Description of the course learning outcomes – CLOs At the end of the course, students will be able to:		PLO	Levels in Learning Domain: Knowledge (Cognitive-C), Attitude (Affective-A), Skills (Psychomotor-P)		
			C	A	P
CLO1	**Produce** exploratory data reports and well-structured, cleaned datasets suitable for predictive modelling tasks.	PLO10	C3		
CLO2	**Construct** modular and reusable machine learning pipelines to automate data processing and model workflows.	PLO3			P4
CLO3	**Justify** the selection of appropriate models and evaluation metrics based on problem requirements and performance results.	PLO2	C5		
CLO4	**Organize** a collaborative project using professional tools, demonstrating effective teamwork and communication.	PLO5		A4	

***** Levels in Learning Domain: Knowledge (Cognitive-C), Attitude (Affective-A), Skills (Psychomotor-P)

Cognitive

1	Remembering	C1
2	Understanding	C2
3	Applying	C3
4	Analyzing	C4
5	Evaluating	C5
6	Creating	C6

Affective

1	Receiving	A1
2	Responding	A2
3	Valuing	A3
4	Organizing	A4
5	Internationalizing	A5

Psychomotor

1	Perception	P1
2	Set	P2
3	Guided Response	P3
4	Mechanism	P4
5	Complex over response	P5
6	Adaptation	P6
7	Origination	P7

15. Mapping of the course learning outcomes to the programme Learning Outcome, Teaching Methods and Assessment methods

CLO	Programme Learning Outcomes – Total Hours for Student Learning Time (SLT) including learning and assessment									
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO 9	PLO10
CLO1										42
CLO2			24							
CLO3		30								

CLO4					24					

CLO	Programme Learning Outcomes – Percentages									
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO 9	PLO10
CLO1										M (35%)
CLO2			P (20%)							
CLO3		P (25%)								
CLO4					P (20%)					

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- **Fully (F)** indicates a focus of more than 50% of the total SLT on this PLO,
- **Moderate (M)** indicates a focus of 31% **-** 50% of the total SLT, and
- **Partial (P)** indicates a focus of less than 30% of the total SLT on the PLO.

CLO	PLO	C/A/P Level	Teaching Method	Assessment Methods
CLO1	PLO10	C3	Lecture + Guided Hands-on Lab + Demonstration (Python, EDA)	Assignment 1 (EDA & Data Cleaning), Mid-term Quiz
CLO2	PLO3	P4	Lab-based Learning + Step-by-step Coding + Scaffolded Exercises (Pipeline building)	Assignment 2 (ML Pipeline Development)
CLO3	PLO2	C5	Case Study + Model Comparison + Discussion + Problem-based Learning	Final Report & Kaggle Submission (Model Justification & Evaluation)
CLO4	PLO5	A4	Team-based Learning + Project-based Learning + Presentation Coaching	Presentation & Defence (Team Project)

16. Distribution of Student Learning Time (SLT)

* Lecture (L), Tutoring (T), Practice (P), Other (O)

	Course Content Outline and subtopics	CLOs	Learning and Teaching Activities									Total SLT
			Face to Face (F2F)								NF2F Independent Learning **(Asynchronous)**	
			Physical				Online/Technology-mediated **(Synchronous)**					
			L	T	P	O	L	T	P	O		
1	**Topic 1:** Intro to Predictive Analytics	CLO1	2	1	1						2	6
2	**Topic 2:** Python for Data Science	CLO1	2		2						2	6
3	**Topic 3:** Exploratory Data Analysis (EDA)	CLO1	2		2						2	6
4	**Topic 4:** Data Cleaning & Preprocessing	CLO1	2		2						2	6
5	**Topic 5:** Baseline Model Development	CLO2	2	1	1						2	6
6	**Topic 6:** Feature Engineering	CLO2	2		2						2	6
7	**Topic 7:** Machine Learning Pipeline	CLO2	2	1	1						2	6
8	**Topic 8:** Regression Models	CLO3	2	1	1						2	6
9	**Topic 9:** Classification Models	CLO3	2		2						2	6
10	**Topic 10:** Model Evaluation	CLO3	2	1	1						2	6
11	**Topic 11:** Model Tuning & Hyperparameters	CLO3	2	1	1						2	6
12	**Topic 12:** Team Project Planning	CLO4	2		2						2	6
13	**Topic 13:** Team Model Development & Comparison	CLO4	2	1	1						2	6
14	**Topic 14:** Data Storytelling & Presentation	CLO4	1	1	2						2	6
											Total SLT for Course Content	84

	Continuous Assessment	%	Face to Face (F2F)		NF2F Independent Learning **(Asynchronous)**	
			Physical	Online/Technology-mediated **(Synchronous)**		
1	Assignment 1	15			8	8
2	Mid-term Quiz	20	2		8	10
3	Assignment 2	20			6	6
4						
5						
					Total SLT for Continuous Assessment:	24
	Final Assessment	%	Face to Face (F2F)		NF2F	

			Physical	Online/Technology-mediated **(Synchronous)**	Independent Learning **(Asynchronous)**	
1	Final Report and Kaggle Submission	25			6	6
2	Presentation & Defence	20	4		2	6
					Total SLT for Final Assessment:	12
					Total SLT for Assessment	36
					Grand Total SLT	120

17. Course Assessment Plan

CLO	PLO	Assessment	Group (G) / Individual (I)	Weight %	SLT	C/A/P Level	Topic															Total Weight (%)	Total SLT
							1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
CLO1	PLO10	Assignment 1	I	15	8	C3	✓	✓	✓	✓												35	18
		Mid-term Quiz	I	20	10		✓	✓	✓	✓													
CLO2	PLO3	Assignment 2	I	20	6	P4					✓	✓	✓									20	6
CLO3	PLO2	Final Report & Kaggle Submission	G	25	6	C5								✓	✓	✓	✓					25	6
CLO4	PLO5	Presentation & Defence	G	20	6	A4												✓	✓	✓		20	6
				100																		100	36

18. Course Outline/detailed lesson plan

Lesson Learning Outcome (LLOs)

Topic		CLOs	Lesson Learning Outcomes (LLOs)
1	**Topic 1:** Intro to Predictive Analytics	CLO1	LLO1: Differentiate supervised vs unsupervised learning. LLO2: Explain ML workflow. LLO3: Setup Python environment for data science.
2	**Topic 2:** Python for Data Science (NumPy & Pandas)	CLO1	LLO1: Apply NumPy operations. LLO2: Use Pandas to explore datasets. LLO3: Configure Kaggle and download datasets.
3	**Topic 3:** Exploratory Data Analysis (EDA)	CLO1	LLO1: Detect missing values and outliers. LLO2: Visualize distributions and correlations.

4	**Topic 4:** Data Cleaning & Preprocessing	CLO1	LLO1: Handle missing values. LLO2: Encode categorical variables. LLO3: Split dataset into train/test sets.
5	**Topic 5:** Baseline Model Development	CLO2	LLO1: Build initial ML models. LLO2: Evaluate model performance.
6	**Topic 6:** Feature Engineering	CLO2	LLO1: Transform and scale features. LLO2: Generate new features.
7	**Topic 7:** Machine Learning Pipeline	CLO2	LLO1: Build modular ML workflows. LLO2: Combine preprocessing and modelling. LLO3: Reuse pipelines for experiments.
8	**Topic 8:** Regression Models	CLO3	LLO1: Train regression models. LLO2: Evaluate using RMSE/MAE.
9	**Topic 9:** Classification Models	CLO3	LLO1: Train classification models. LLO2: Evaluate Accuracy/F1/ROC-AUC.
10	**Topic 10:** Model Evaluation	CLO3	LLO1: Analyse model performance metrics. LLO2: Visualize confusion matrices/ROC curves.
11	**Topic 11:** Model Tuning & Hyperparameters	CLO3	LLO1: Optimize models with GridSearchCV. LLO2: Improve metrics; balance bias-variance trade-off.
12	**Topic 12:** Team Project Planning	CLO4	LLO1: Define dataset, KPIs, and team roles. LLO2: Use Notion/Trello to plan workflow.
13	**Topic 13:** Team Model Development & Comparison	CLO4	LLO1: Build collaborative pipelines LLO2: Compare models; generate leaderboard-ready submission
14	**Topic 14:** Data Storytelling & Presentation	CLO4	LLO1: Communicate model results LLO2: Explain business impact
15			

* Active Learning Strategies (ALS)

Detail Course Syllabus

Week	Hour (L/T/P/O)	Lecture	CLOs	Teaching Method/Activity	Learning Method/Activity (ALS)	Assessment (Weight %)	T&L Resources
1	2/1/1/0	**Topic 1**: Intro to Predictive Analytics - LLO1, LLO2 and LLO3	CLO1	Lecture, Demo	Listening, Guided exploration	0	Lecture Slides, Dataset
2	2/0/2/0	**Topic 2**: Python for Data Science - LLO1, LLO2 and LLO3	CLO1	Lecture, Coding Lab	Hands-on practice	0	Jupyter, Python
3	2/0/2/0	**Topic 3**: Exploratory Data Analysis (EDA) - LLO1, LLO2 and LLO3	CLO1	Lecture, Lab	Data exploration	0	Lecture slides, Pandas, Visualization
4	2/0/2/0	**Topic 4**: Data Cleaning & Preprocessing - LLO1 and LLO2	CLO1	Lecture, Lab	Problem-solving	0	Dataset

5	2/1/1/0	Assignment 1 **Topic 5**: Baseline Model Development - LLO1 and LLO2	CLO2	Lecture, Guided Lab	Practice	**15 (Assignment 1)**	Scikit-learn, Project Phase 1
6	2/0/2/0	**Topic 6**: Feature Engineering - LLO1 and LLO2	CLO2	Lecture, Lab	Feature creation	0	Dataset
7	2/1/1/0	**Topic 7:** Machine Learning Pipeline - LLO1, LLO2 and LLO3	CLO2	Lecture, Demo	Guided implementation	0	sklearn. pipeline
8	0/0/2/0	Midterm Exam	CLO1	Quiz Exam	Individual work	**20**	Exam paper
9	2/1/1/0	**Topic 8:** Regression Models - LLO1 and LLO2	CLO3	Lecture, Lab	Practice	0	Lab
10	2/0/2/0	**Topic 9:** Classification Models - LLO1 and LLO2	CLO3	Lecture, Lab	Practice	0	Dataset
11	2/1/1/0	**Topic 10:** Model Evaluation - LLO1 and LLO2	CLO3	Lecture, Case Study	Analysis	0	Metrics tools
12	2/1/1/0	**Topic 11:** Model Tuning & Hyperparameters - LLO1 and LLO2	CLO3	Lecture, Lab	Experimentation	**20 (Assignment 2)**	GridSearchCV, Project Progress
13	2/0/2/0	**Topic 12:** Team Project Planning - LLO1 and LLO2	CLO4	Lecture	Collaboration	0	Tracking tools
14	2/1/1/0	**Topic 13:** Model Development & Comparison - LLO1 and LLO2	CLO4	Team Lab	Group work	0	Coding tools
15	2/1/0/1	**Topic 14**: Data Storytelling & Presentation - LLO1 and LLO2	CLO4	Feedback	Communication	**25 (Final Report)**	Slides, Final Project
16	0/0/0/4	**Final Exam: Presentation (Defence)**	CLO4	Final Presentation	Team	**20**	Defence, Q&A
						100	

19. Require resources to deliver the course (E.g., Pix4D mapper, Green seeker, Weather Station, Computer lab, etc.)

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20. Required References/Textbooks and Other Teaching and Learning materials

Required references/Textbooks

Title	Author(s)	Publisher	Year	ISBN
An Introduction to Statistical Learning: with Applications in Python	Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani	Springer	2023 (2nd Edition)	978-3031387467

Recommended Textbooks/ Readings

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd Edition)	Aurélien Géron	O'Reilly Media	2023 (3rd Edition)	978-1098125974

Other Teaching and learning materials:

- Kaggle – datasets, notebooks, and competitions for hands-on practice

21. Student Responsibility

- Read the learning materials beforehand which means to be ready for the next session.
- Be completely engaged in class, active, take notes, and ask questions in class.
- Students are required to participate in various activities in and out of the classroom, such as group work, and individual homework for assignments and reports assigned by the subject teacher.
- Submit the assigned tasks such as homework, assignments, or research proposal on time.

22. Rubric and Rating scale

CLOs	PLO	C/A/P Level	Assessment and description	Weight (%)	Evaluation Definition	Scoring
CLO1	PLO10	C3	Assignment 1 (EDA & Data Cleaning)	15	Ability to apply EDA techniques and perform proper data cleaning and preprocessing	0-15
			Mid-term Quiz	20	Ability to apply predictive analytics concepts and interpret data-related problems	0-20
CLO2	PLO3	P4	Assignment 2 (ML Pipeline Development)	15	Ability to construct and implement a functional machine learning pipeline (data → model → output)	0–15
CLO3	PLO2	C5	Final Report & Kaggle Submission	25	Ability to evaluate and justify model selection using appropriate metrics, comparison, and reasoning	0–25

CLO4	PLO5	A4	Presentation & Defence	25	Ability to organize teamwork, communicate results clearly, and defend decisions professionally	0–25
Total Weightage (%)				**100**		**100**

23. Course policy

Students must abide by the following policies:

Attendance & Preparation

- Students are expected to attend all classes (online/on-site).
- Assigned readings and preparation outlines must be completed before class if any.
- Active participation in group discussions and post-chapter evaluations is required.

Academic Integrity

- All assignments, quizzes, and exams must be done individually.
- Academic misconduct includes:
  - Presenting others' work as one's own.
  - Engaging in plagiarism or cheating.

Homework & Assignments

- Preparation outlines must be submitted on time; late submissions incur penalties.
- Peer discussion is encouraged, but midterm and final exams must be independent.

Examinations

- Midterm and final exams are closed-book.

Penalties

- Violations of academic integrity may result in a failing grade and referral to the Office of Student Affairs.
- Students uncertain about potential violations should consult the instructor in advance.

24. Rating Scale

Letter Grade	Grade Point	Score	Explanation
A	4.00	85-100	Excellent
B+	3.50	80-84	Very Good
B	3.00	75-79	Good
C+	2.50	70-74	Fairly Good
C	2.00	65-69	Fair
D+	1.50	60-64	Poor
D	1.00	50-59	Very Poor
F	0.00	<50	Fail

25. Date

January 2026